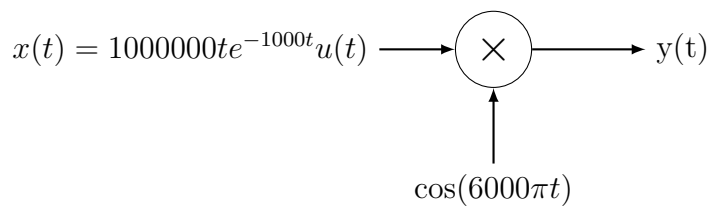
ECE 3710 – Quiz 3 – LTI Systems

Name: Solution Work On Last Page →

T#: _____

Consider the following Fourier Transform pairs, and the system shown below with input $x(t) = 1000000te^{-1000t}u(t)$ and output $y(t) = x(t)\cos(6000\pi t)$. Answer the questions below. You may use the back, but clearly indicate your answers.

Time Domain	Frequency Domain
$te^{-at}u(t)$	$\dfrac{1}{(a+j2\pi f)^2}$
$g(t)\cos(2\pi f_c t)$	$\dfrac{1}{2}[G(f-f_c)+G(f+f_c)]$

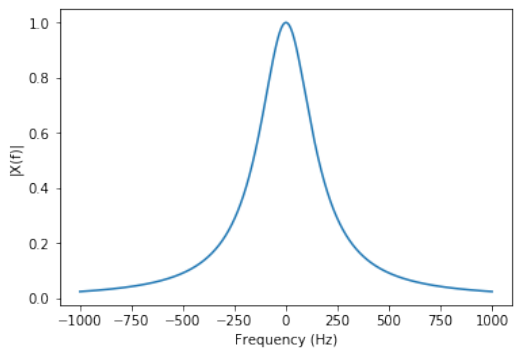


1. (3 points) Find the Fourier Transform of the input, $X(f)$.

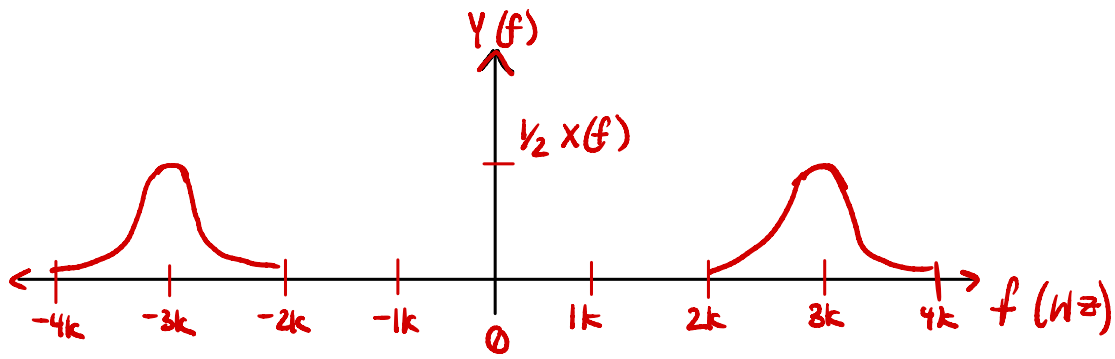
$$\boxed{X(f) = \frac{1{,}000{,}000}{(1{,}000 + j2\pi f)^2}}$$

2. (3 points) Find the Fourier Transform of the output, $Y(f)$.

$$\boxed{Y(f) = \frac{500{,}000}{(1{,}000 + j2\pi(f - 3{,}000))^2} + \frac{500{,}000}{(1{,}000 + j2\pi(f + 3{,}000))^2}}$$



3. (3 points) The magnitude spectrum of the input, $|X(f)|$ is plotted above. (For simplicity, assume that the magnitude is zero outside of the plot.) Sketch the magnitude spectrum of the output, $|Y(f)|$. Label the axes.



$$Y(f) = \frac{500{,}000}{(1{,}000 + j2\pi(f - 3{,}000))^2} + \frac{500{,}000}{(1{,}000 + j2\pi(f + 3{,}000))^2}$$

Problem ①:

$$x(t) = 1{,}000{,}000\, e^{-1{,}000t}\, u(t)$$

$$\boxed{X(f) = \frac{1{,}000{,}000}{(1{,}000 + j2\pi f)^2}}$$

Time Domain	Frequency Domain
$te^{-at}u(t)$	$\frac{1}{(a+j2\pi f)^2}$
$g(t)\cos(2\pi f_c t)$	$\frac{1}{2}[G(f-f_c)+G(f+f_c)]$

Problem ②:

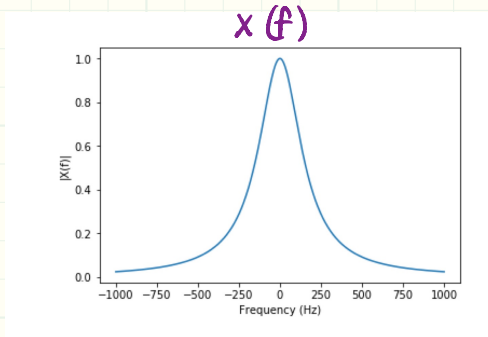
$$x(t) = 1{,}000{,}000\, e^{-1{,}000t}\, u(t) \qquad y(t) = x(t)\cos(6{,}000\pi t)$$

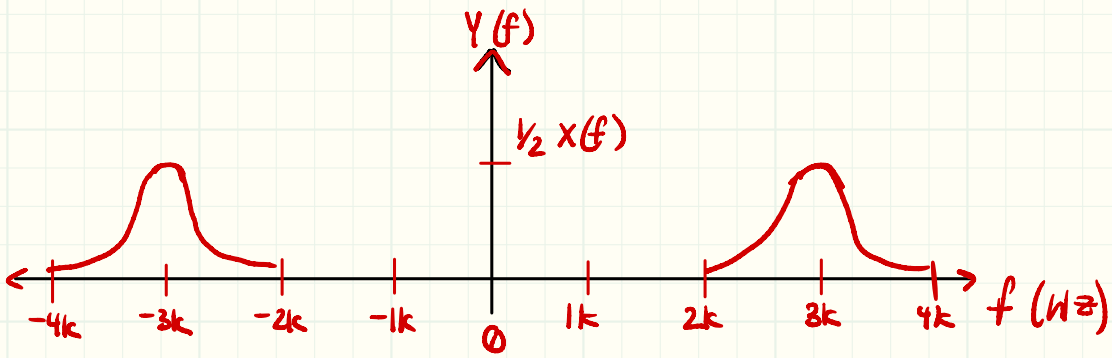
$$Y(f) = \frac{1}{2}\left[X(f-f_c) + X(f+f_c)\right] \quad ; \quad \omega = 2\pi f \rightarrow f = \frac{6{,}000\pi}{2\pi}$$

$$f = 3{,}000 \text{ Hz}$$

$$\boxed{Y(f) = \frac{500{,}000}{(1{,}000 + j2\pi(f-3{,}000))^2} + \frac{500{,}000}{(1{,}000 + j2\pi(f+3{,}000))^2}}$$

Problem ③:





$$\boxed{Y(f) = \frac{500{,}000}{(1{,}000 + j2\pi(f-3{,}000))^2} + \frac{500{,}000}{(1{,}000 + j2\pi(f+3{,}000))^2}}$$